

# Week 2 Recitation

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*ECON8711 (Micro 1A)*

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# TA Introduction

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  - Feel free to reach out!
  - Please indicate in the subject field: "[ECON8711] ~ "  
➔ This would help me get to you quickly.
- Recitation: Fri 1:00 - 2:50PM, Ramseyer 115
  - 50 min talk - 10 min break - 50 min talk
  - Team work!
- Office hour: By appointment, Zoom option available

# Production Technology

How to represent what a firm produces

- Production Possibility Set:  $Z \in \mathbb{R}^K$ 
  - $K$  materials a firm can use & generate
  - $y_k^i$ : input of material  $k$
  - $y_k^o$ : output of material  $k$   
 $\Rightarrow z_k = y_k^o - y_k^i$  : net-put
- Input Requirement Set

$$V(y) = \{x \in \mathbb{R}_+^I \mid (y, -x) \in Z\}$$

- Isoquant for a single output  $y \in \mathbb{R}_+$

$$Q(y) = \{x \in \mathbb{R}_+^I \mid x \in V(y), \text{ and for } \forall y' > y, x \notin V(y')\}$$

# Production Technology

## Question in class

Suppose you have production function  $f : \mathbb{R}_+^I \rightarrow \mathbb{R}_+^O$ .

Write down (1) Production possibility set; (2) Input requirement set; (3) Isoquant.

# Production Technology

## Question in class

Suppose you have production function

$$f(x_1, x_2) = A \min\{a_1 x_1, a_2 x_2\} \text{ with } a_1, a_2 > 0.$$

Write down (1) Production possibility set; (2) Input requirement set;  
(3) Isoquant.

# Production Technology

## Another example

Suppose there are two goods productions of  $(y_1, y_2) \in \mathbb{R}_+^2$  by inputs  $(x_1, x_2, x_3) \in \mathbb{R}_+^3$ . Consider the production function

$$f(x_1, x_2) = (2x_1 + 4x_2, \min\{x_2, 2x_3\}).$$

When factor prices are  $w = (w_1, w_2, w_3) = (3, 2, 4)$ , graph the set of outputs that costs less than 20. (i.e.,

$$\{(y_1, y_2) \in \mathbb{R}_+^2 \mid w \cdot x \leq 20 \text{ and } y = f(x)\}.)$$

# Production Technology

## Useful conditions of Prod Tech

- Monotonic Technology

- Monotonic Input Requirement Sets:

If  $x \in V(y)$  and  $x' \geq x$ , then  $x' \in V(y)$

- Monotonic Production Possibility set:

If  $z \in Z$  and  $z' \leq z$ , then  $z' \in Z$

- Convex Technology

- Convexity on  $V(y)$ :

If  $x, x' \in V(y)$ , then for  $\forall \lambda \in [0, 1]$ ,  $\lambda x + (1 - \lambda)x' \in V(y)$

- Convexity on  $Z$ :

If  $z, z' \in Z$ , then for  $\forall \lambda \in [0, 1]$ ,  $\lambda z + (1 - \lambda)z' \in Z$

- Regular Technology

$V(y)$  is closed and non-empty

# Function and Set convexity

- Set convexity:

A set  $C \in \mathbb{R}^N$  is convex when for all  $x, y \in C$  and for all  $\lambda \in [0, 1]$

$$\lambda x + (1 - \lambda)y \in C$$

- Function Convexity:

A function  $f : \mathbb{R}^N \rightarrow \mathbb{R}$  is convex when the set

$$\{(x, r) \in \mathbb{R}^{N+1} | r \geq f(x)\}$$

is convex.

- Function Quasi-Convexity: Domain Version

A function  $f : \mathbb{R}^N \rightarrow \mathbb{R}$  is quasi-convex when the set

$$S_\alpha(f) = \{x \in \mathbb{R}^N | f(x) \leq \alpha\}$$

is convex.



# Function and Set convexity

## Properties of convexity

- Quasi-convexity as a generalization of Convexity

### Proposition

Let  $f : \mathcal{D} \rightarrow \mathbb{R}$  be a concave function. If  $f$  is concave on  $\mathcal{D}$ ,  $f$  is also quasi-concave on  $\mathcal{D}$ . If  $f$  is convex on  $\mathcal{D}$ ,  $f$  is also quasi-convex on  $\mathcal{D}$ .

# Function and Set convexity

## Properties of convexity

- Every concave or convex function must also be continuous on the interior of its domain.

### Proposition

Let  $f : \mathcal{D} \rightarrow \mathbb{R}$  be a concave function. If  $\mathcal{D}$  is open,  $f$  is continuous on  $\mathcal{D}$ . If  $\mathcal{D}$  is not open,  $f$  is continuous on the interior of  $\mathcal{D}$ .

# Production Technology

## Some auxiliary conditions of Prod Tech

- Technology Rate of Substitution (TRS)  
E.g., TRS of Cobb-Douglas production function  
 $f(x_1, x_2) = Ax_1^a x_2^{(1-a)}$ ?

- Elasticity

$$\epsilon = \frac{\frac{dy}{y}}{\frac{dx}{x}} = \frac{dy}{dx} \frac{x}{y} = \frac{d \ln y}{d \ln x}$$

# Production Technology

## Some auxiliary conditions of Prod Tech

- Homogeneous of Degree  $k$

$$f(tx) = t^k f(x)$$

- Homothetic function

$$f(tx) = g(h(x))$$

where  $h$  is Homogeneous of Degree 1 and  $g$  is a monotonic function.